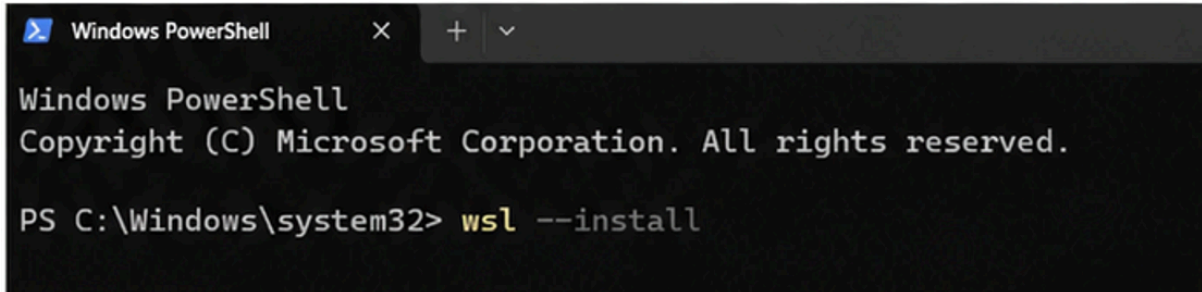


WSL INSTALLATION

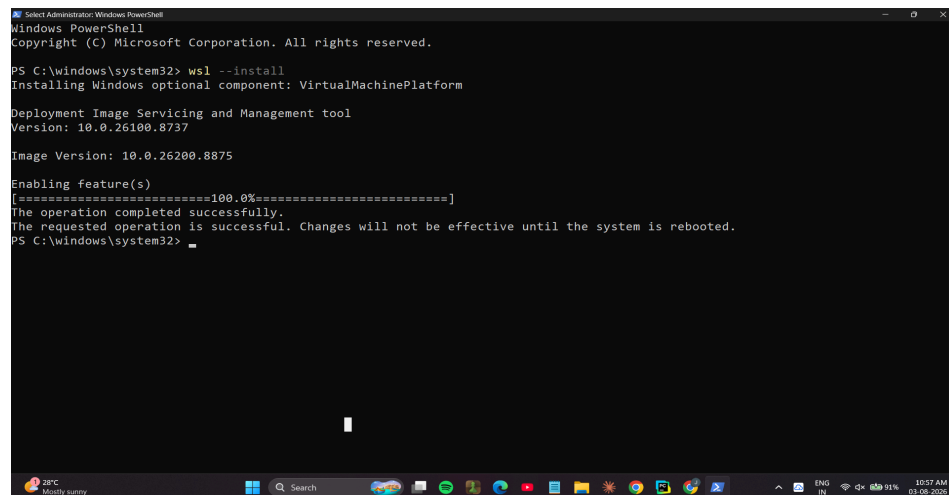
1. PowerShell opened as Administrator:



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Windows\system32> wsl --install
```

2. Run:
`wsl --install`



```
Select Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Windows\system32> wsl --install
Installing Windows optional component: VirtualMachinePlatform
Deployment Image Servicing and Management tool
Version: 10.0.26100.8737
Image Version: 10.0.26200.8875
Enabling feature(s)
[=====100.0%=====]
The operation completed successfully.
The requested operation is successful. Changes will not be effective until the system is rebooted.
PS C:\Windows\system32>
```

3. Restart Your Computer

After installation, restart your PC when prompted.

4. Complete Ubuntu Setup

After restarting:

- =>Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
- =>Wait for the installation to finish.
- =>Create a Linux username.

Example:

Enter new UNIX username: anusha

5. Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password:****

6. Update Ubuntu

```
jhanasri_thota@DESKTOP-7F3G2K1: ~  
jhanasri_thota@DESKTOP-7F3G2K1: ~$ sudo apt update  
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease  
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease  
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease  
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease  
Hit:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages  
Hit:6 http://archive.ubuntu.com/ubuntu noble/universe Translation-en  
Hit:7 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages  
Hit:8 http://archive.ubuntu.com/ubuntu noble/multiverse Translation-en  
Hit:9 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages  
Hit:10 http://archive.ubuntu.com/ubuntu noble/restricted Translation-en  
Hit:11 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages  
Hit:12 http://archive.ubuntu.com/ubuntu noble/main Translation-en  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
All packages are up to date.
```

7. Install GCC (C Compiler)

```
jhanasri_thota@DESKTOP-7F3G2K1: ~$ sudo apt install build-essential  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
build-essential is already the newest version (12.9ubuntu3).  
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

8. Verify the Installation

```
jhanasri_thota@DESKTOP-7F3G2K1: ~$ gcc --version  
gcc (Ubuntu 13.2.0-23ubuntu4) 13.2.0  
Copyright (C) 2023 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

9. Create Your C Program - ex:nano process.cc

10. Compile the Program

```
jhanasri_thota@DESKTOP-7F3G2K1: ~$ gcc process.c -o process  
$ ./process
```

SIMPLE COMMANDS:

Topic 2: Directory Management

```
jhanasri@LAPTOP-1RN1M8E3:~$ ls
Desktop Documents Downloads Pictures src

jhanasri@LAPTOP-1RN1M8E3:~$ ls -l
total 20
drwxr-xr-x 2 jhanasri jhanasri 4096 Aug  8 13:40 Desktop
drwxr-xr-x 2 jhanasri jhanasri 4096 Aug  8 13:41 Documents
drwxr-xr-x 2 jhanasri jhanasri 4096 Aug  8 13:41 Downloads
drwxr-xr-x 2 jhanasri jhanasri 4096 Aug  8 13:41 Pictures
drwxr-xr-x 2 jhanasri jhanasri 4096 Aug  8 13:42 src

jhanasri@LAPTOP-1RN1M8E3:~$ ls -a
.  ..  .bash_history .bash_logout .bashrc .cache Desktop
Documents Downloads Pictures .profile .sudo_admin_successf
src
jhanasri@LAPTOP-1RN1M8E3:~$ cd Documents
jhanasri@LAPTOP-1RN1M8E3:~/Documents$ pwd
/home/jhanasri/Documents
jhanasri@LAPTOP-1RN1M8E3:~/Documents$ cd ..
jhanasri@LAPTOP-1RN1M8E3:~$ mkdir OSLab
jhanasri@LAPTOP-1RN1M8E3:~$ ls
Desktop Documents Downloads OSLab Pictures src
jhanasri@LAPTOP-1RN1M8E3:~$ rmdir OSLab
jhanasri@LAPTOP-1RN1M8E3:~$ find . -name "*.c"
./src/tssk1.c
./src/task1.c
./src/task2.c
jhanasri@LAPTOP-1RN1M8E3:~$ locate hello.c
/usr/include/boost/icl/detail/hello.hpp
/home/jhanasri/src/hello.c
jhanasri@LAPTOP-1RN1M8E3:~$
```

Topic 3: File Management

```
jhanasri@LAPTOP-1RN1M8E3:~$ touch file1.txt
jhanasri@LAPTOP-1RN1M8E3:~$ ls
Desktop Documents Downloads file1.txt Pictures src
jhanasri@LAPTOP-1RN1M8E3:~$ cat file1.txt
jhanasri@LAPTOP-1RN1M8E3:~$ echo "Hello Linux" > file1.txt
jhanasri@LAPTOP-1RN1M8E3:~$ cat file1.txt
Hello Linux
jhanasri@LAPTOP-1RN1M8E3:~$ cp file1.txt copy.txt
jhanasri@LAPTOP-1RN1M8E3:~$ ls
copy.txt Desktop Documents Downloads file1.txt Pictures src
jhanasri@LAPTOP-1RN1M8E3:~$ mv copy.txt newfile.txt
jhanasri@LAPTOP-1RN1M8E3:~$ ls
Desktop Documents Downloads file1.txt newfile.txt Pictures src
jhanasri@LAPTOP-1RN1M8E3:~$ rm newfile.txt
jhanasri@LAPTOP-1RN1M8E3:~$ ls
Desktop Documents Downloads file1.txt Pictures src
jhanasri@LAPTOP-1RN1M8E3:~$ head -n 1 /etc/passwd
root:x:0:0:root:/root:/bin/bash
jhanasri@LAPTOP-1RN1M8E3:~$ tail -n 1 /etc/passwd
systemd-coredump:x:999:999:systemd Core Dumper:/usr/sbin/nologin
jhanasri@LAPTOP-1RN1M8E3:~$
```

Topic 4: File Editing

```
jhanasri@LAPTOP-1RN1M8E3:~$ nano hello.c
# nano editor opened)
# Created a simple C file
# Save: Ctrl+O, Enter
# Exit: Ctrl+X

jhanasri@LAPTOP-1RN1M8E3:~$ cat hello.c
#include <stdio.h>
int main()
{
    printf("Hello Linux\n");
    return 0;
}

jhanasri@LAPTOP-1RN1M8E3:~$ vi hello.c
# (vi editor opened)
# Press i to insert
# Save: Esc then :wq
# Exit

jhanasri@LAPTOP-1RN1M8E3:~$ █
```

Topic 5: C Program Compilation and Execution

```
jhanasri@LAPTOP-1RN1M8E3:~$ gcc hello.c -o hello
jhanasri@LAPTOP-1RN1M8E3:~$ ls
Desktop  Documents  Downloads  hello.  file1.txt  Pictures  src
jhanasri@LAPTOP-1RN1M8E3:~$ ./hello
Hello Linux
jhanasri@LAPTOP-1RN1M8E3:~$ which gcc
/usr/bin/gcc
jhanasri@LAPTOP-1RN1M8E3:~$ whereis gcc
gcc: /usr/bin/gcc /usr/include/gcc /usr/share/man/man1/gcc.1.gz
jhanasri@LAPTOP-1RN1M8E3:~$ man gcc | head -n 5
GCC(1)                General Commands Manual                GCC(1)

NAME
    gcc - GNU project C and C++ compiler

SYNOPSIS
    gcc [options] file...

jhanasri@LAPTOP-1RN1M8E3:~$ █
```

Topic 6: Process Management


```
jhanasri@LAPTOP-1RN1M8E3:~$ ps
  PID TTY          TIME CMD
  976 pts/1        00:00:00 bash
 1563 pts/1        00:00:00 ps

jhanasri@LAPTOP-1RN1M8E3:~$ ps -ef | head -n 5
UID          PID    PPID  C STIME TTY          TIME CMD
jhanasri    976     975  0 13:30 pts/1        00:00:00 bash
jhanasri   1564     976  0 13:31 pts/1        00:00:00 ps
root         1         0  0 12:15 ?            00:00:00 init
root         9         2  0 12:15 ?            00:00:00 [kthreadd]

jhanasri@LAPTOP-1RN1M8E3:~$ top -b -n 1 | head -n 8
top - 13:31:55 up 1:16, 0 users, load average: 0.00, 0.01, 0.05
Tasks: 11 total, 1 running, 10 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.2 us, 0.4 sy, 0.0 ni, 98.4 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
Mem Mem : 3863.4 total, 1252.0 free, 966.1 used, 1651.3 buff/cache
Mem Swap: 2048.0 total, 2048.0 free, 0.0 used, 2537.7 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  MEM     TIME+ COMMAND
  976 jhanasri  20   0 31836 2048 2708  S   0.0   0.1   0:00.02 bash
 1569 jhanasri  20   0 45448 2316 2304  S   0.0   0.1   0:00.01 top

jhanasri@LAPTOP-1RN1M8E3:~$ sleep 1000 &
[1] 1581

jhanasri@LAPTOP-1RN1M8E3:~$ ps | grep sleep
1581 pts/1        00:00:00 sleep
jhanasri@LAPTOP-1RN1M8E3:~$
jhanasri@LAPTOP-1RN1M8E3:~$
```

Topic 7: Job Control

```
jhanasri@LAPTOP-1RN1M8E3:~$ sleep 1000 &
[1] 1605

jhanasri@LAPTOP-1RN1M8E3:~$ sleep 1000 &
[2] 1606

jhanasri@LAPTOP-1RN1M8E3:~$ jobs
[1]-  Running                  sleep 1000 &
[2]+  Running                  sleep 1000 &

jhanasri@LAPTOP-1RN1M8E3:~$ bg %1
[1]+  sleep

jhanasri@LAPTOP-1RN1M8E3:~$ fg %2
sleep 1000
^C
[2]+  Terminated              sleep 1000

jhanasri@LAPTOP-1RN1M8E3:~$ ./hello &
[2] 1632

jhanasri@LAPTOP-1RN1M8E3:~$ jobs
[1]-  Running                  sleep 1000 &
[2]-  Running                  ./hello &

jhanasri@LAPTOP-1RN1M8E3:~$
```

Topic 8: File Permissions and Links

```

jhanasri@LAPTOP-1RN1M8E3:~$ touch script.sh
jhanasri@LAPTOP-1RN1M8E3:~$ ls -l script.sh
-rw-r--r-- 1 jhanasri jhanasri 0 Aug  8 13:40 script.sh
jhanasri@LAPTOP-1RN1M8E3:~$ chmod +x script.sh
jhanasri@LAPTOP-1RN1M8E3:~$ ls -l script.sh
-rwxr-xr-x 1 jhanasri jhanasri 0 Aug  8 13:40 script.sh
jhanasri@LAPTOP-1RN1M8E3:~$ chmod 755 script.sh
jhanasri@LAPTOP-1RN1M8E3:~$ ls -l script.sh
-rwxr-xr-x 1 jhanasri jhanasri 0 Aug  8 13:40 script.sh
jhanasri@LAPTOP-1RN1M8E3:~$ touch file1
jhanasri@LAPTOP-1RN1M8E3:~$ ln file1 file2
jhanasri@LAPTOP-1RN1M8E3:~$ ls -li file1 file2
131072 -rw-r--r-- 2 jhanasri jhanasri 0 Aug  8 13:42 file1
131072 -rw-r--r-- 2 jhanasri jhanasri 0 Aug  8 13:42 file2
jhanasri@LAPTOP-1RN1M8E3:~$ ln -s file1 shortcut
jhanasri@LAPTOP-1RN1M8E3:~$ ls -l shortcut
lrwxrwxrwx 1 jhanasri jhanasri 5 Aug  8 13:42 shortcut -> file1

```

Topic 9: Disk and Memory Information

```

jhanasri@LAPTOP-1RN1M8E3:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb         250G   42G   196G  18% /
tmpfs            2.0G   1.7M   2.0G   1% /dev/shm
tmpfs            2.0G   1.1M   2.0G   1% /run
tmpfs            5.0M     0   5.0M   0% /run/lock
jhanasri@LAPTOP-1RN1M8E3:~$ du -sh .
120K  .
jhanasri@LAPTOP-1RN1M8E3:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.8Gi        1.1Gi        1.2Gi         17Mi        1.5Gi        2.4Gi
Swap:          2.0Gi          0B          2.0Gi
jhanasri@LAPTOP-1RN1M8E3:~$ █

```

Topic 10: User and Environment Management

```
jhanasri@LAPTOP-1RN1MBE3:~$ id
uid=1000(jhanasri) gid=1000(jhanasri) groups=1000(jhanasri),27(sudo),121(docker)
jhanasri@LAPTOP-1RN1MBE3:~$ env | head -n 5
SHELL=/bin/bash
PWD=/home/jhanasri
LOGNAME=jhanasri
XDG_SESSION_TYPE=tty
HOME=/home/jhanasri
jhanasri@LAPTOP-1RN1MBE3:~$ export MYVAR=HelloLinux
jhanasri@LAPTOP-1RN1MBE3:~$ echo $MYVAR
HelloLinux
jhanasri@LAPTOP-1RN1MBE3:~$ alias ll='ls -l'
jhanasri@LAPTOP-1RN1MBE3:~$ ll | head -n 3
total 36
-rw-r--r-- 1 jhanasri jhanasri      0  Aug  8 13:42 file1
-rw-r--r-- 2 jhanasri jhanasri      0  Aug  8 13:42 file2
jhanasri@LAPTOP-1RN1MBE3:~$
```